Instagram Data Analysis Report

Notebook Link – Instagram_Data_Analysis

GitHub Repository - Task-03_instagram_data_analysis

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1. Introduction -

Social media platforms have become one of the most powerful tools for digital marketing and audience engagement. Instagram is widely used by brands and businesses to increase visibility, improve engagement, and grow followers.

This project focuses on analyzing Instagram engagement data to understand:

- Best posting times for maximum engagement
- Impact of content types and filters on engagement
- Popular hashtag usage patterns
- Factors influencing audience interaction

The objective of this analysis is to extract meaningful insights and provide data-driven recommendations that can help improve social media strategies for platforms like **Alfido Tech**.

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2. Dataset Description -

Source – Kaggle

Link - https://www.kaggle.com/datasets/bhanupratapbiswas/instgram

The dataset contains Instagram-related information including:

- Instagram post details
- Posting dates and times
- Filters used on posts
- Content/photo types
- Likes and comments data
- Hashtag information
- User interaction records

This dataset is useful for analyzing audience engagement behavior and identifying effective content strategies.

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3. Methodology -

Data Cleaning -

Data preprocessing was performed to improve data quality and consistency. The following operations were completed:

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Loaded multiple Instagram-related datasets

- Parsed date columns into proper datetime format
- Extracted posting hour and posting day from timestamps
- Merged likes and comments data with photo data
- Handled missing values using appropriate techniques
- Encoded categorical variables using LabelEncoder

Engagement Metrics -

Important engagement metrics were calculated including:

- Likes per post
- Comments per post
- Total engagement score

The engagement score was computed using:

Engagement Score = Total Likes + Total Comments

These metrics helped evaluate post-performance and audience interaction.

Exploratory Data Analysis (EDA) -

EDA was performed to understand engagement patterns and posting behavior:

- Posting frequency by hour was analyzed
- Posting frequency by day was examined
- Content type distribution was studied
- Instagram filter usage patterns were analyzed
- Popular hashtags were identified
- Engagement trends across posts were explored

These analyses helped uncover trends related to user engagement and content performance.

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4. Visualizations -

The following visualizations were created:

Posting Frequency by Hour Chart

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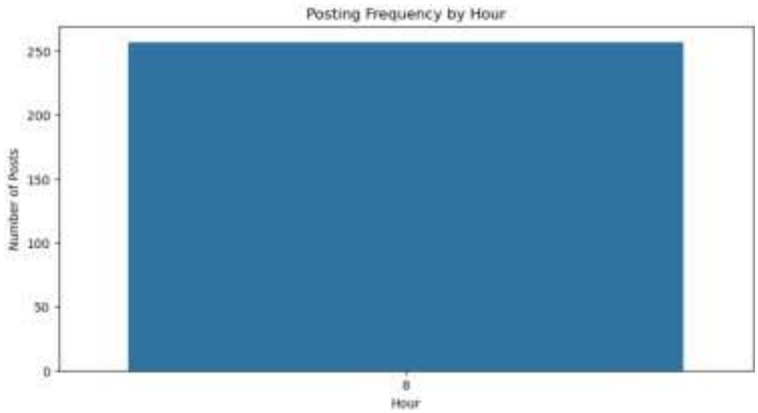
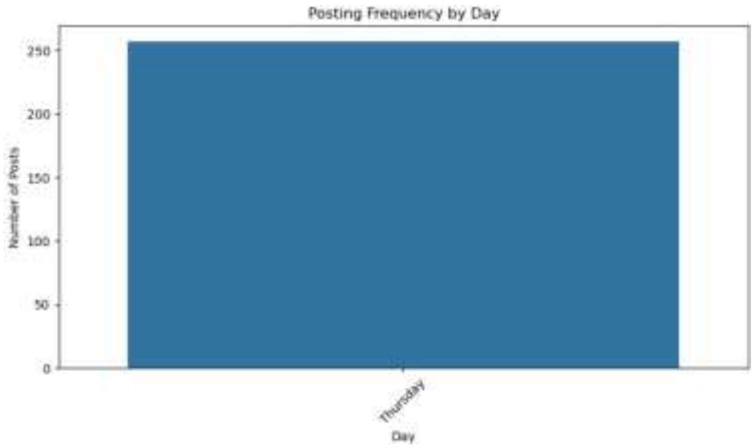
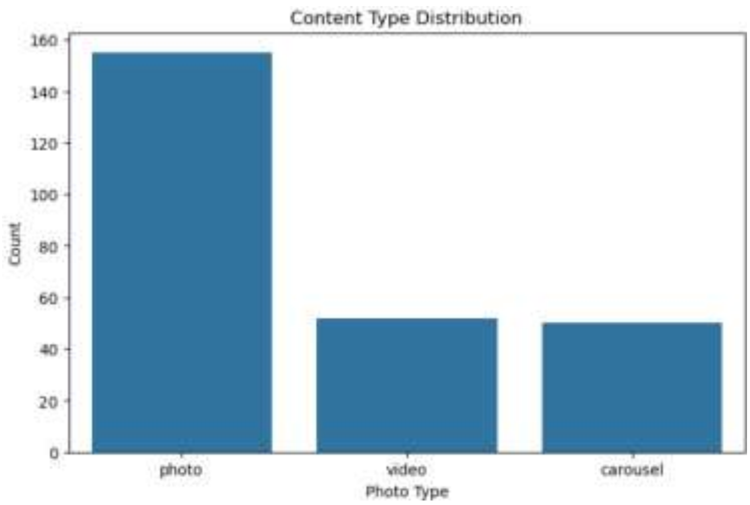


• Posting Frequency by Day Chart



• Content Type Distribution Plot



Top Instagram Filters Used Chart

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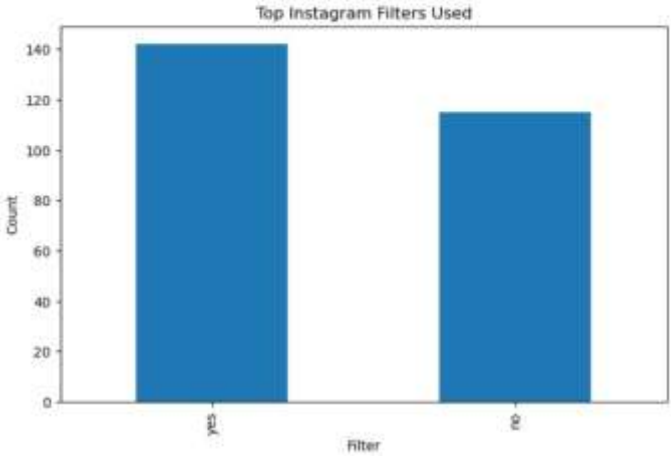
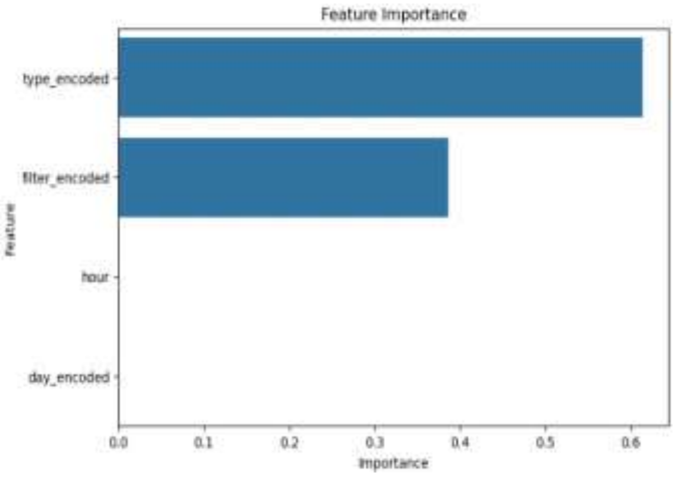


- Feature Importance Chart



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5. Machine Learning Model -

To further enhance the analysis, a machine learning model was developed to predict Instagram engagement scores. **Model Used -**

Random Forest Regressor -

Random Forest Regressor was selected because:

- It handles non-linear relationships effectively
- It works well with mixed feature types
- It reduces overfitting compared to single decision trees
- It provides reliable feature importance analysis

Features Used -

The following features were used as input variables:

- Posting hour
- Posting day
- Instagram filter used
- Content/photo type **Target Variable:**
- Engagement Score

Model Pipeline -

A machine learning workflow was implemented including:

- Feature encoding using Label-Encoder
- Train-test data splitting
- Random Forest Regressor model training
- Prediction and performance evaluation

This pipeline ensures efficient preprocessing and consistent model training.

Model Performance

The model was evaluated using the following metrics:

- MAE (Mean Absolute Error)
- RMSE (Root Mean Squared Error)
- R^2 Score

Interpretation:

- Lower MAE and RMSE indicate better prediction accuracy
- Higher R^2 Score indicates stronger model performance

The model successfully captured engagement trends and posting behavior patterns.

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6. Key Findings -

The analysis revealed several important insights:

- Certain posting hours generate higher engagement
- Consistent posting schedules improve visibility
- Specific content types perform better than others
- Some Instagram filters are associated with higher engagement
- Strategic hashtag usage improves audience reach and interaction

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7. Recommendations (Alfido Tech Platform) -

Based on the analysis, the following recommendations are proposed:

- Post content during peak engagement hours
- Maintain a consistent posting schedule
- Use visually engaging content formats
- Utilize trending and niche hashtags strategically
- Use high-performing filters and content styles consistently

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8. Optimal Content Calendar - Suggested

Posting Strategy:

- Post during high-engagement hours
- Focus on weekdays with better interaction
- Mix informative and visually engaging content
- Maintain regular audience interaction through comments and replies
- Use hashtag optimization to increase visibility

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9. Conclusion -

This project successfully analyzed Instagram engagement data and identified important trends related to posting schedules, content strategies, and audience interaction.

The study shows that posting time, content type, hashtags, and filters significantly influence engagement performance.

The integration of machine learning further strengthens the analysis by enabling prediction of engagement scores using historical posting behavior.

These insights can help platforms like **Alfido Tech** improve social media strategies, optimize content planning, and increase audience engagement.

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